

DANIEL SZUREK

+1 (205) 213-6132 | [Email](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)
Tuscaloosa, AL, USA

EDUCATION

The University of Alabama

Master of Science in Computer Science
Accelerated Master's Program (AMP)

May 2025 - December 2026

GPA: 3.67

The University of Alabama

Bachelor of Science in Computer Science
Minor in Mathematics, Honors College, Upsilon Pi Epsilon

August 2022 - May 2026

GPA: 3.87

PROFESSIONAL EXPERIENCE

General Motors

ADAS Software Engineer Intern

Milford, MI, USA

May 2026 - August 2026

- Designing and developing an LLM-powered coding assistance agent aimed to automate repetitive function creation for the data recording team, with the goal of reducing function creation time by 80%.

The University of Alabama EcoCAR

Connected and Automated Vehicle (CAV) Lead

Tuscaloosa, AL, USA

September 2025 - Present

- Architected the autonomous navigation stack for a pre-production Cadillac Lyriq, which integrated Lane Centering, Cooperative Adaptive Cruise Control, and V2X connectivity for a robust Advanced Driver Assistance System (ADAS) that communicated with stock onboard systems
- Developed longitudinal control algorithms (Model Predictive Control/PID) in MATLAB/Simulink, a lateral Stanley controller, and a global nearest neighbor (GNN) algorithm for sensor fusion
- Validated system performance through rigorous Hardware-in-the-Loop (HIL) and Vehicle-in-the-Loop (VIL) testing, ensuring safety-critical reliability
- Conducted research on gaussian processes, MPCs, and reinforcement learning for autonomous systems
- Led a team of 15 students in the final year of the EcoCAR EV Challenge, achieving first in vehicle technical specifications, second in MathWorks Model-Based Design, Automatic Intersection Navigation, and CAV Presentations, and third in automatic intersection navigation energy consumption

SSAB

Data Science Co-Op

Mobile, AL, USA

May 2025 - August 2025

- Engineered automated ETL pipelines using Azure Data Factory and SQL to process large-scale manufacturing datasets.
- Designed and implemented a project portfolio dashboard in Power BI, automating quarterly financial reporting
- Built and deployed cost-comparison solutions in Tableau and SQL
- Simplified workflows for high-volume datasets
- Built dynamic production capacity models using Power BI, enabling leadership to define optimal targets based on live sensor data
- Led presentations for senior managers and hosted international meetings with colleagues in Sweden and Finland

PROJECTS

Driveability Accessible Driving Resource

[Link to Project](#)

- Developed a social media website with resources and reviews on assisted driving tools as part of my degree's capstone project
- Aimed at disabled users aiming to learn or spread information for accessible driving

Gaussian Process and MPC Controller for Automatic Intersection Navigation via V2X

[Link to Project](#)

- Developed a custom controller for Automatic Intersection Navigation that accounts for SPaT V2X data loss from radio noise via a trained Gaussian Process

Super-Resolution Facial Recognition Pipeline

[Link to Project](#)

- Developed a custom Convolutional Neural Network (CNN) pipeline to perform facial recognition on low-resolution inputs
- Implemented a Deep Super-Resolution (DSR) model to upscale images before feeding them into a quantized EdgeFace model, improving accuracy on degraded inputs
- Applied model quantization techniques to reduce model size for efficient deployment without significant accuracy loss

Budgie: Smart Expense Planning

[Link to Project](#)

- Built a budgeting application for iOS in Swift with sporadic student income in mind
- Features a custom constraint-based scheduling algorithm that predicts the most optimal date to purchase wish list items based on income and expenses.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, MATLAB, SQL, JavaScript (React), Dart (Flutter), Java, Swift, Scala, Lisp, Haskell, Lox, Assembly

AI & Robotics: ROS 2, Model Predictive Control (MPC), Computer Vision (CNNs), Sensor Fusion, Simulink, RTMaps, Gaussian Processes, Reinforcement Learning, PyTorch, LangChain, LLM Integration, machine learning, embedded software & systems

Cloud & DevOps: Azure (Data Factory, Databricks), Docker, Git, Linux (Ubuntu), JSON

Web & Mobile: Full-Stack Development, REST APIs, Flutter, React, Backend Engineering, Frontend Engineering, UI